



Solution Brief

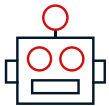
Cloudflare + Ping Identity: Securing Model Context Protocol (MPC) Servers for the Modern Enterprise

Easily secure your MCP servers with Ping
Identity and Cloudflare Workers

Challenge

AI agent frameworks are rapidly becoming core infrastructure for modern enterprises — automating workflows, querying internal systems, and acting on behalf of users. Model Context Protocol (MCP) has emerged as the standard way for AI clients to connect to APIs and tools.

But this new access pattern introduces security gaps that traditional controls were not built to address:



AI agents are unauthenticated by default.

Most MCP deployments lack a standardized mechanism to verify which agent is connecting, on whose behalf, and with what level of authorization.



Client onboarding is manual and fragmented.

Registering AI clients with an identity provider requires IT intervention, creating provisioning bottlenecks and shadow agent risk.



Downstream API access is uncontrolled.

When an MCP server calls internal APIs on behalf of a user, those calls often carry no user context, bypassing fine-grained authorization policies entirely.



Compliance requirements are unmet.

Regulated industries need verifiable proof that AI access was authorized by an authenticated user, under defined scopes, with a clear audit trail.

Solution

Together, Cloudflare Workers and Ping Identity make enterprise identity the foundation of every MCP interaction. MCP servers are deployed as Cloudflare Workers at the edge, with Ping Identity serving as the central authorization authority for every agent connection, user authentication, and downstream API call.

Dynamic Client Registration (DCR) enables AI clients to onboard automatically with Ping Identity at connection time — no manual provisioning required. **User authentication and consent** are enforced through the enterprise identity provider, ensuring agents only act after a user has explicitly authenticated and approved. Token Exchange (RFC 8693) then allows the MCP server to call downstream APIs using a delegated token scoped specifically to the target resource — carrying the user's identity, bound to the session, and limited to only the permissions required.

Every API call made by an AI agent is traceable back to a specific authenticated user, a specific MCP session, and a specific set of granted permissions.

Supported Ping Identity Platforms PingOne

Uses the Cloudflare Workers OAuth Provider to act as a standards-compliant OIDC client to PingOne. Cloudflare manages the consent dialog; PingOne handles authentication. Ideal for organizations on PingOne Multi-Tenant looking for the fastest path to a secure MCP deployment.

PingOne Advanced Identity Cloud (AIC)

Implements metadata-driven DCR with RFC 8414 discovery and a two-stage Token Exchange — the MCP server obtains an actor token via client credentials, then performs a delegated exchange producing a token with the `act` claim (RFC 8693) scoped to the target API.

PingFederate

Mirrors the AIC pattern using PingFederate's authorization server, with a single-stage Token Exchange scoped to the target API resource. Designed for enterprises running PingFederate on-premises or in private cloud environments.

Why Does Identity Matter for AI Agents?

AI agents operating through MCP are not passive tools — they read data, write records, and trigger workflows on behalf of users. Without identity controls at every layer, agents become a privileged, unmonitored access path into enterprise systems.

Ping Identity addresses this at the protocol level:

- **Multi-Factor Authentication (MFA):** Users authenticate with MFA before any agent session is established, preventing account takeover from propagating into AI-driven workflows.
- **Single Sign-On (SSO):** Users authenticate once with their existing enterprise credentials. No separate agent credentials to manage or rotate.
- **Dynamic Client Registration:** AI clients are registered automatically and governed by Ping Identity policy — giving security teams a complete registry of authorized agents without manual intervention.
- **Scoped Token Exchange:** Agents only receive tokens for the specific APIs they need, with only the scopes required for the current operation. Tokens cannot be reused outside the session.
- **Delegated Access with `act` Claims:** The full delegation chain is cryptographically encoded in the token — who the user is, which agent acted on their behalf, and under what authorization.
- **Revocation and Session Control:** Tokens are short-lived and session-bound. Revoking access at the identity provider immediately terminates all downstream agent access — no credential rotation required.

About Ping Identity

Ping Identity delivers intelligent identity solutions for the enterprise. They enable companies to achieve Zero Trust identity-defined security and more personalized, streamlined user experiences. The PingOne Cloud Platform provides customers, workforce, and partners with access to cloud, mobile, SaaS, and on-premises applications across the hybrid enterprise. Ping Identity's portfolio spans single sign-on, multi-factor authentication, dynamic authorization, and no-code orchestration — forming the identity foundation for secure AI deployments.

About Cloudflare

Cloudflare, Inc. is on a mission to help build a better Internet. Cloudflare's platform protects and accelerates any Internet application online without adding hardware, installing software, or changing a line of code. Internet properties powered by Cloudflare have all web traffic routed through its intelligent global network, which gets smarter with every request. Cloudflare Workers brings this network to serverless compute — enabling developers to deploy MCP servers, APIs, and AI infrastructure at the edge with built-in security, global distribution, and zero infrastructure management.